

Chronic Disease Management

Long-Term Care Protocols for Major Non-Communicable Diseases

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1. Diabetes Mellitus — Long-Term Management

1.1 Glycaemic Targets

Individualised glycaemic targets balance the benefits of tight glucose control (microvascular complication prevention) against hypoglycaemia risk. Targets should be relaxed in the elderly, those with limited life expectancy, or frequent severe hypoglycaemia.

Target	General Adult	Elderly / Complex	Pregnancy (GDM/Pre-existing DM)
HbA1c	< 7.0% (53 mmol/mol)	< 8.0–8.5%	< 6.5% (pre-existing) / 6.0–6.5% (GDM)
Fasting BGL	4.4–7.2 mmol/L	< 8.3 mmol/L	3.5–5.3 mmol/L
Post-prandial BGL	< 10.0 mmol/L	< 11.1 mmol/L	< 7.8 mmol/L (1-hr) or < 6.7 (2-hr)

1.2 Pharmacotherapy Algorithm (Type 2 DM)

- Step 1: Lifestyle modification (diet, exercise, weight loss) + Metformin (if tolerated, eGFR $\geq$ 30)
- Step 2 (if HbA1c above target): Add SGLT2 inhibitor (CV/renal benefit) OR GLP-1 RA (weight loss) OR DPP-4i
- Step 3: Triple therapy — add 3rd agent from different class based on comorbidities
- Step 4: Initiate basal insulin if HbA1c remains uncontrolled; intensify with bolus insulin as needed
- ASCVD/Heart Failure present: Prioritise SGLT2i (empagliflozin, dapagliflozin) regardless of HbA1c
- Obesity with DM: Prioritise GLP-1 RA (semaglutide, dulaglutide) for weight-glycaemic dual benefit

1.3 Complication Monitoring Schedule

Complication	Screening Test	Frequency
Diabetic nephropathy	Urine albumin-creatinine ratio + eGFR	Annually
Diabetic retinopathy	Dilated funduscopy or retinal photography	Annually (T1DM: 5 yrs post-diagnosis)
Diabetic neuropathy	Monofilament, vibration, ankle reflexes	Annually

Complication	Screening Test	Frequency
Cardiovascular disease	BP, lipid profile, ECG, ankle-brachial index	Annually; 3-monthly BP
Foot care	Full foot examination including pulses	Annually; 3-monthly if high risk
HbA1c	Glycated haemoglobin	Every 3 months (new/poorly controlled) or every 6 months

2. Chronic Heart Failure (CHF)

2.1 Classification & Diagnosis

Heart failure is classified by left ventricular ejection fraction (LVEF): HFrEF (LVEF < 40%), HFmrEF (LVEF 40–49%), and HFpEF (LVEF ≥ 50%). Echocardiography is the cornerstone of diagnosis and guides therapy.

Clinical Features (Framingham Criteria)

- Major: Paroxysmal nocturnal dyspnoea, orthopnoea, elevated JVP, S3 gallop, cardiomegaly, acute pulmonary oedema
- Minor: Ankle oedema, nocturnal cough, exertional dyspnoea, hepatomegaly, pleural effusion, tachycardia (> 120/min)
- Diagnosis: 2 major criteria OR 1 major + 2 minor criteria

2.2 HFrEF (LVEF < 40%) — Disease-Modifying Therapy

Four drug classes form the evidence-based cornerstone of HFrEF management (the 'fantastic four') and have proven mortality benefit:

Drug Class	Example	Target Dose (Titrate)	Key Benefit
ACE inhibitor / ARB	Ramipril / Sacubitril-Valsartan	Ramipril 10 mg BD Sacubitril-Valsartan 200 mg BD	Neurohormonal blockade; reduces mortality and hospitalisation
Beta-blocker	Bisoprolol, Carvedilol	Bisoprolol 10 mg OD	Anti-remodelling; reduces sudden cardiac death
MRA (Aldosterone Antagonist)	Spironolactone, Eplerenone	25–50 mg OD	Additional neurohormonal blockade; reduces mortality
SGLT2 Inhibitor	Dapagliflozin, Empagliflozin	10 mg OD	Reduced HF hospitalisation and CV death; now guideline-recommended

3. Chronic Kidney Disease (CKD)

3.1 Staging by eGFR

Stage	GFR (mL/min/1.73m ²)	Description	Management Focus
G1	≥ 90	Normal or high — if kidney damage markers present	Risk factor modification; monitor
G2	60–89	Mildly decreased	Treat underlying cause; CVD risk reduction
G3a	45–59	Mildly to moderately decreased	Nephrology involvement; anaemia, PTH, lipids
G3b	30–44	Moderately to severely decreased	Medication review; diet; avoid nephrotoxins
G4	15–29	Severely decreased	Renal replacement therapy planning; AV fistula
G5	< 15	Kidney failure (ESKD)	Dialysis / transplantation / palliative care

■ **Avoid in CKD: NSAIDs (reduce GFR, cause Na/water retention), aminoglycosides, IV contrast (nephrotoxic), metformin (if eGFR < 30 — lactic acidosis risk), high potassium supplements, direct oral anticoagulants in G5.**

3.2 CKD Complication Management

- Anaemia: Erythropoiesis-stimulating agents (ESA) when Hb < 100 g/L; iron supplementation to maintain TSAT ≥ 20%
- Metabolic acidosis: Sodium bicarbonate supplementation if bicarbonate < 22 mmol/L
- CKD-MBD: Phosphate restriction, phosphate binders, Vitamin D analogues (calcitriol), cinacalcet for secondary hyperparathyroidism
- Hyperkalaemia: Dietary potassium restriction; review medications; patiromer or sodium zirconium cyclosilicate for chronic hyperkalaemia
- Hypertension: Target BP < 130/80 mmHg; ACEi/ARB first-line (slow CKD progression, reduce proteinuria)
- Proteinuria: ACEi/ARB to reduce to < 0.5 g/day where possible; SGLT2i if diabetic CKD

4. Chronic Obstructive Pulmonary Disease (COPD)

4.1 GOLD Severity Classification (Spirometry)

GOLD Grade	FEV1 % predicted (post-bronchodilator)	Severity	Symptom Burden
1	≥ 80%	Mild	Low (mMRC 0–1, CAT < 10)
2	50–79%	Moderate	Low or High (mMRC ≥2, CAT ≥10)
3	30–49%	Severe	High — frequent exacerbations (≥2/yr or 1 hospitalisation)
4	< 30%	Very Severe	Very High — multiple exacerbations; respiratory failure

4.2 Pharmacological Management

- Short-acting bronchodilators (SABA/SAMA): Salbutamol, ipratropium — for acute symptom relief (all patients)
- Long-acting bronchodilators (LABA/LAMA): Formoterol, tiotropium — maintenance therapy (GOLD 2–4)
- Dual bronchodilation (LABA+LAMA): First-line for high-symptom or frequent exacerbation patients
- Inhaled corticosteroids (ICS): Add if eosinophil count ≥ 300 cells/μL or frequent exacerbations despite dual bronchodilation
- Triple therapy (ICS+LABA+LAMA): For patients with high eosinophils and frequent exacerbations
- Smoking cessation: Most effective intervention to slow FEV1 decline — pharmacotherapy + counselling
- Pulmonary rehabilitation: Reduces dyspnoea, improves exercise tolerance and quality of life
- Long-term oxygen therapy (LTOT): If PaO₂ ≤ 7.3 kPa (or 7.3–8.0 kPa with cor pulmonale/polycythaemia); ≥15 hours/day

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